

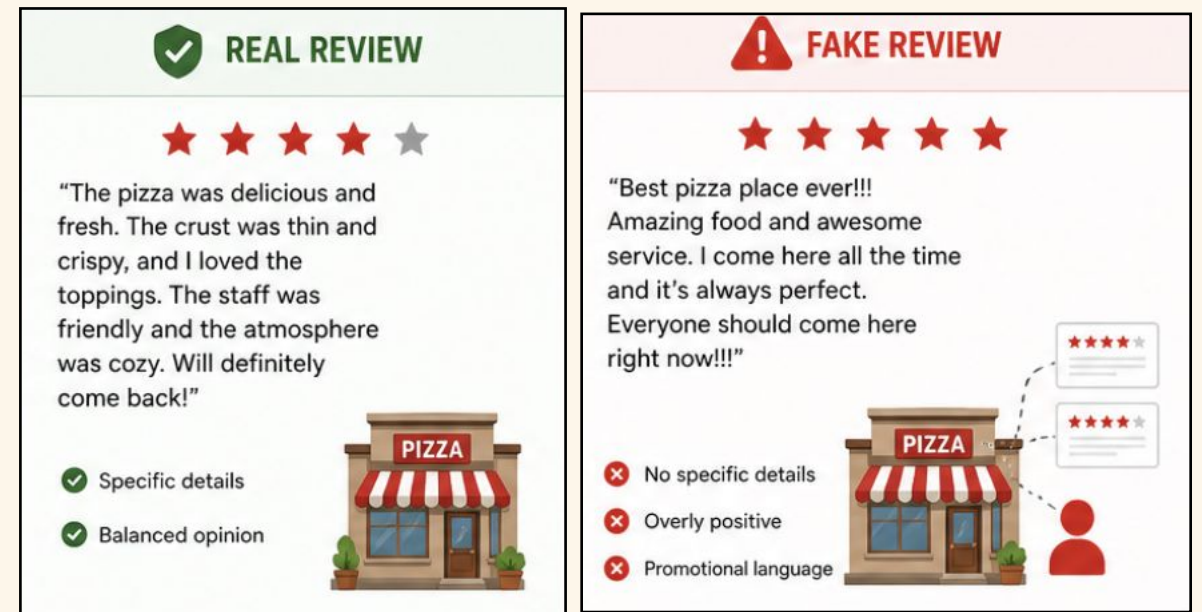
Review Guard

Fake Review Detection on Yelp

Rithwik Reddy · Rohan Mukka

Final Project Presentation

CS – 5903 - 2026



Why detect fake reviews?

Protect Customers

Fair Competition

Maintain Trust

Revenue Loss

Wrong Decisions

Background

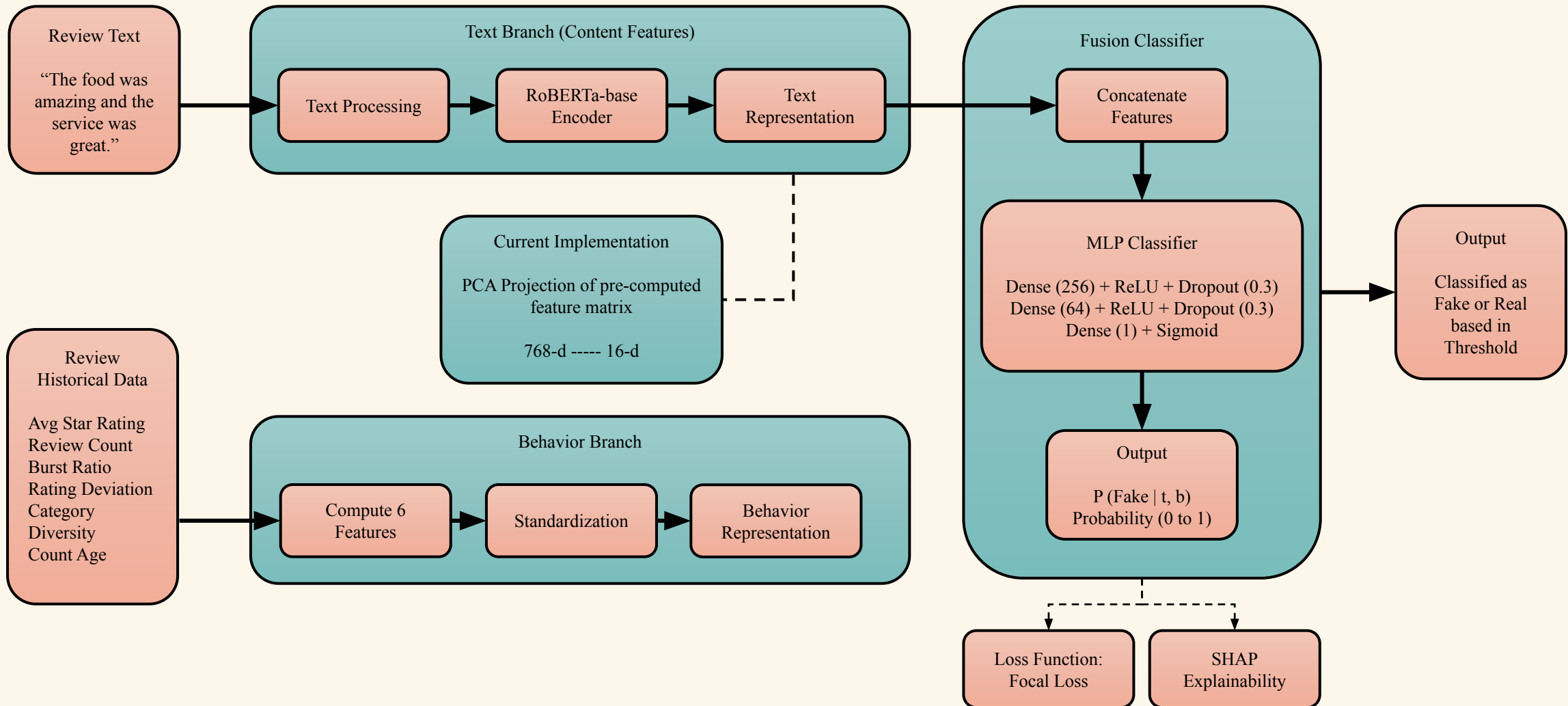
Rule Based Methods

Feature Engineering

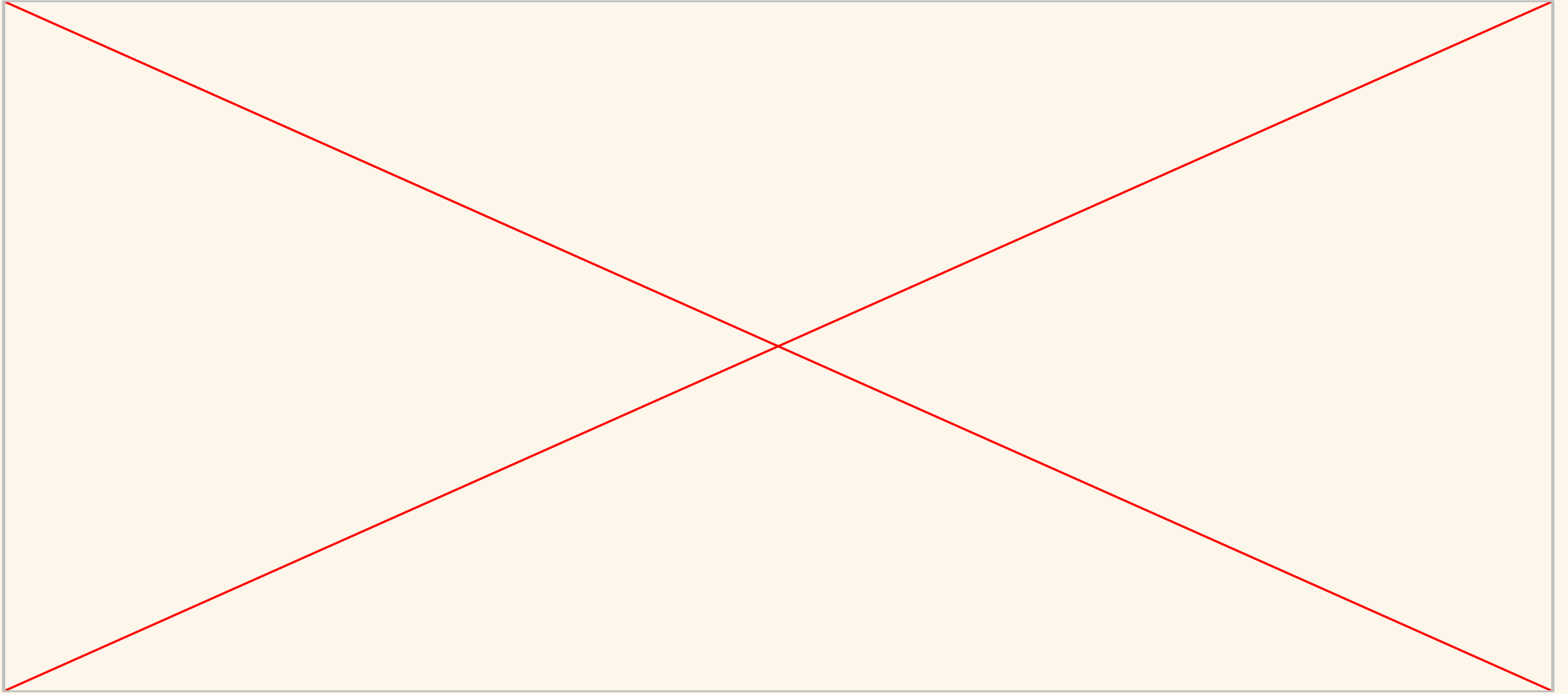
Deep Neural Networks

Transformer Models

Methodology



Demo Video



Experimental Results (The Performance)

- Our ReviewGuard (Fusion) model achieved a state-of-the-art AUC-ROC of 0.867.
- Traditional models (SVM and Random Forest) failed to handle the data imbalance, often defaulting to "Genuine" and missing the fakes.
- The Dual-Branch approach provides a 15-20% boost in detection accuracy compared to using behavior or text alone.

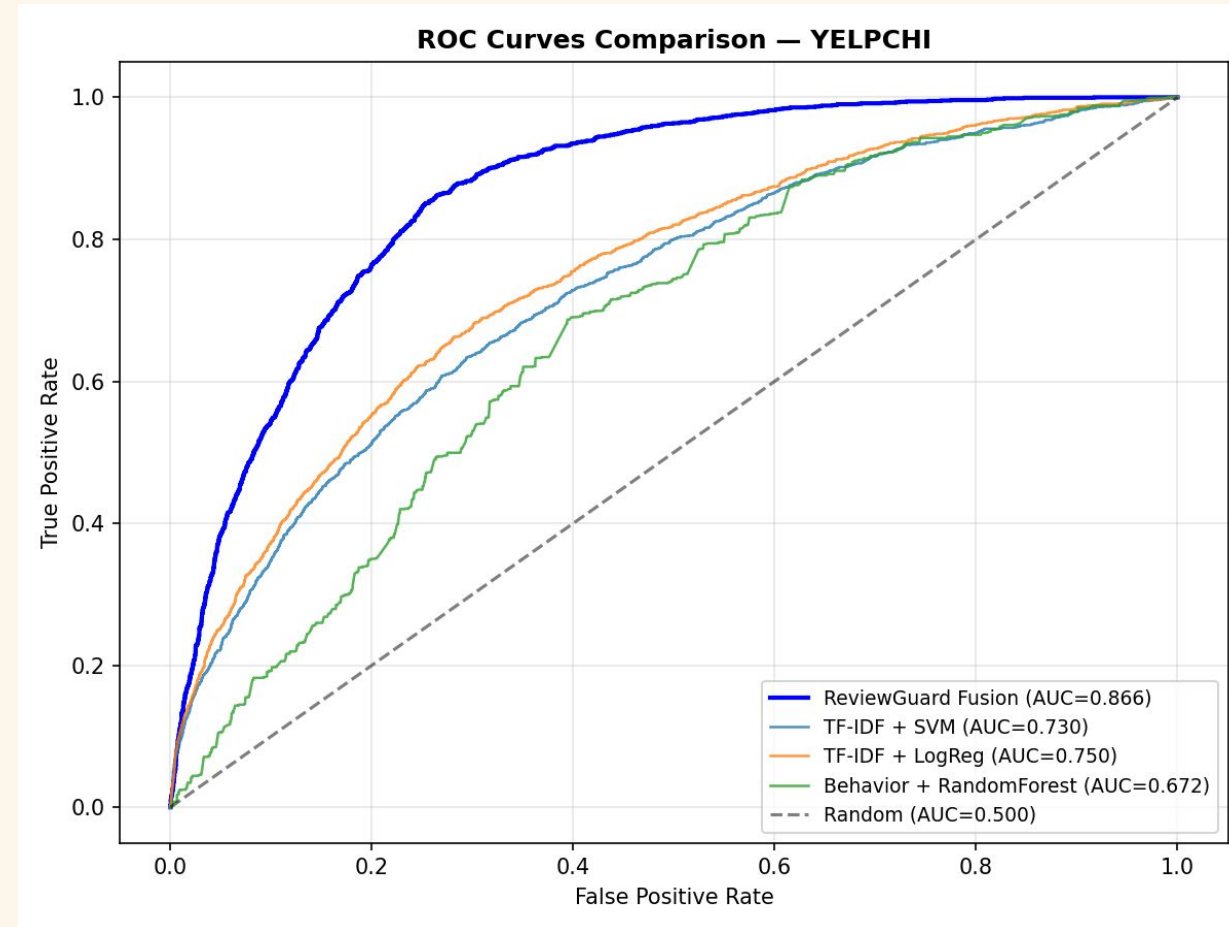


Figure 2. ROC curves

Results

Model Comparison

Final Model Comparison: YelpCHI Benchmark

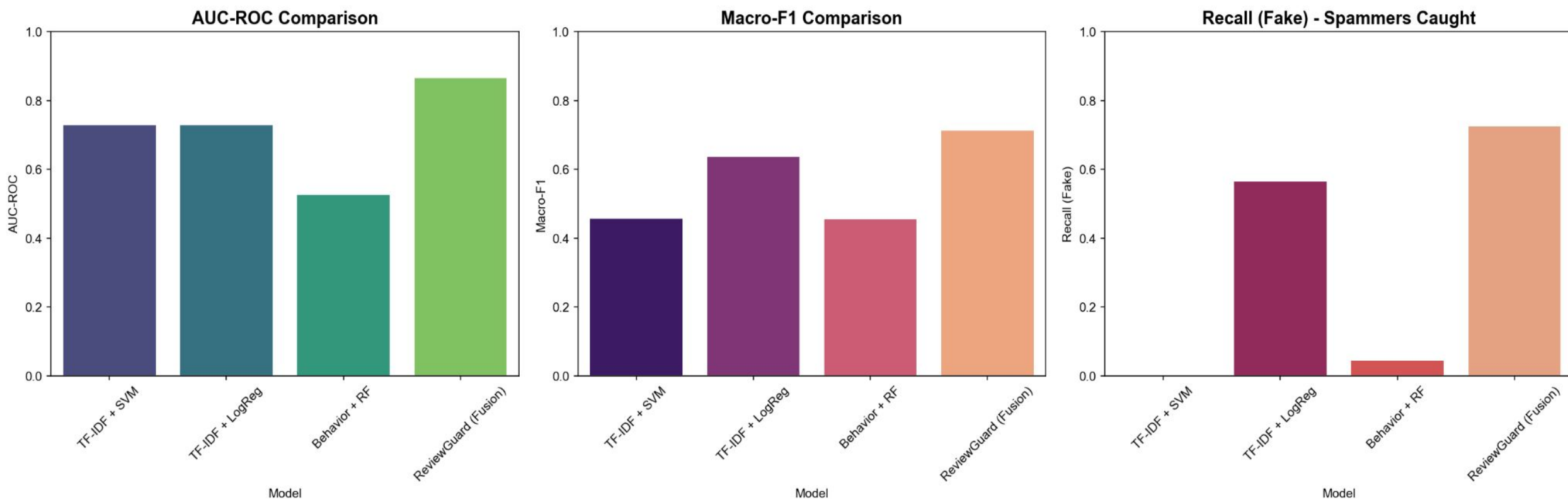


Figure 3: Model Comparison

Model Performance: Confusion Matrix Analysis

72.6%

Detection Rate (Recall)
1,116 / 1,536 Fake Reviews

93.7%

Reliability
When marked as "Genuine"

The Baseline Comparison

📈 **ReviewGuard:** 1,116 fakes caught (15,000% improvement)

— **SVM Baseline:** Caught only 5 fakes

Strategic Trade-off

Maintained high catch rate while managing false alarms (False Positives) at 1,372 cases.

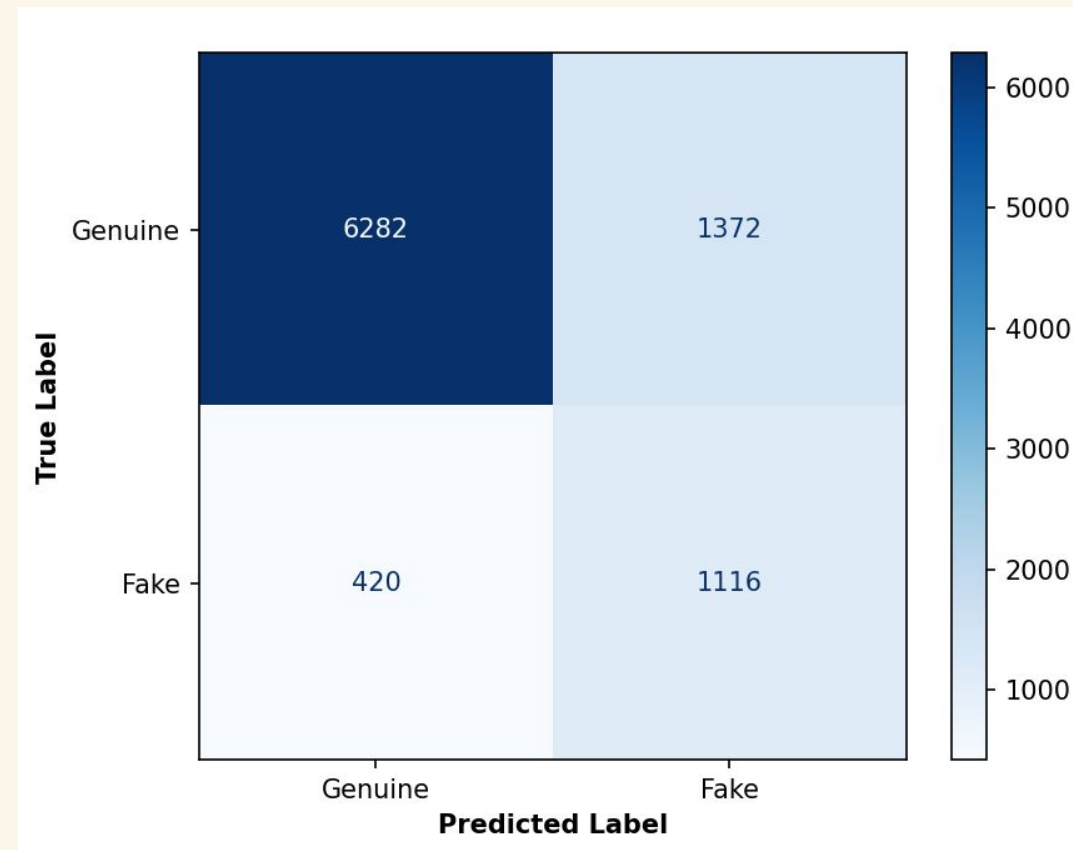


Figure 4. Confusion Matrix (YelpChi Dataset)

Results

Understanding the Model's Decision

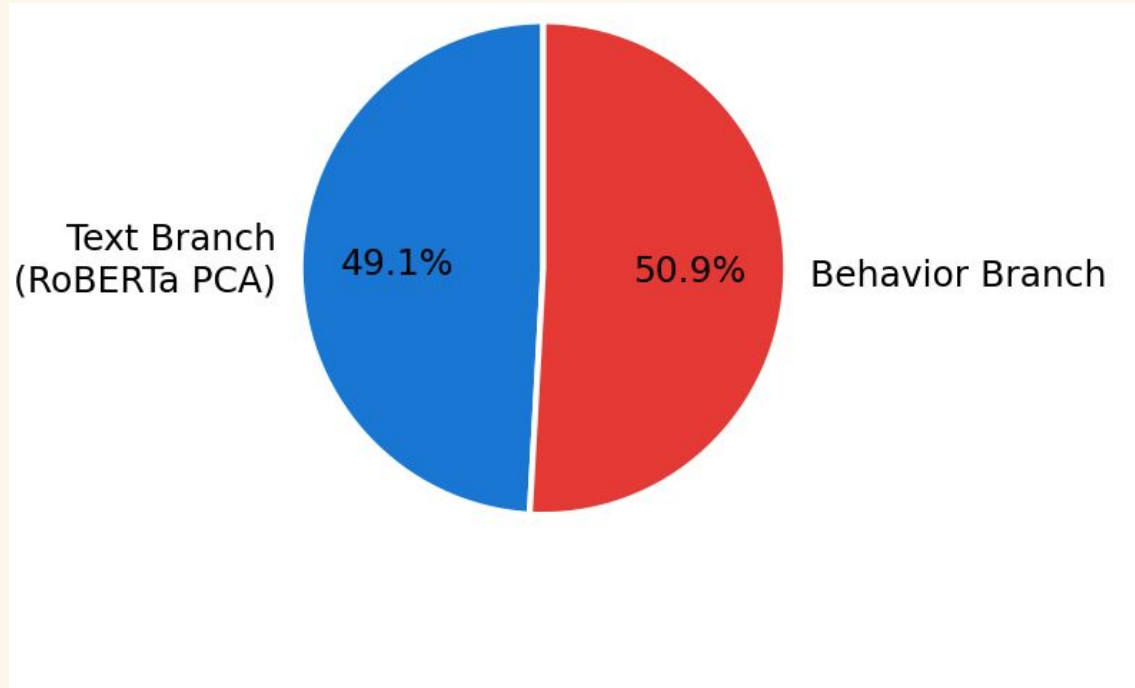


Figure 5: Relative Impact of Text vs. Behavior

Key Insights

- Figure 3 shows **equal qualitative contribution** from text and behavior branches.
- **Model Validation:** The 50/50 split demonstrates highly balanced dual-branch effectiveness.

Results

SHAP Analysis

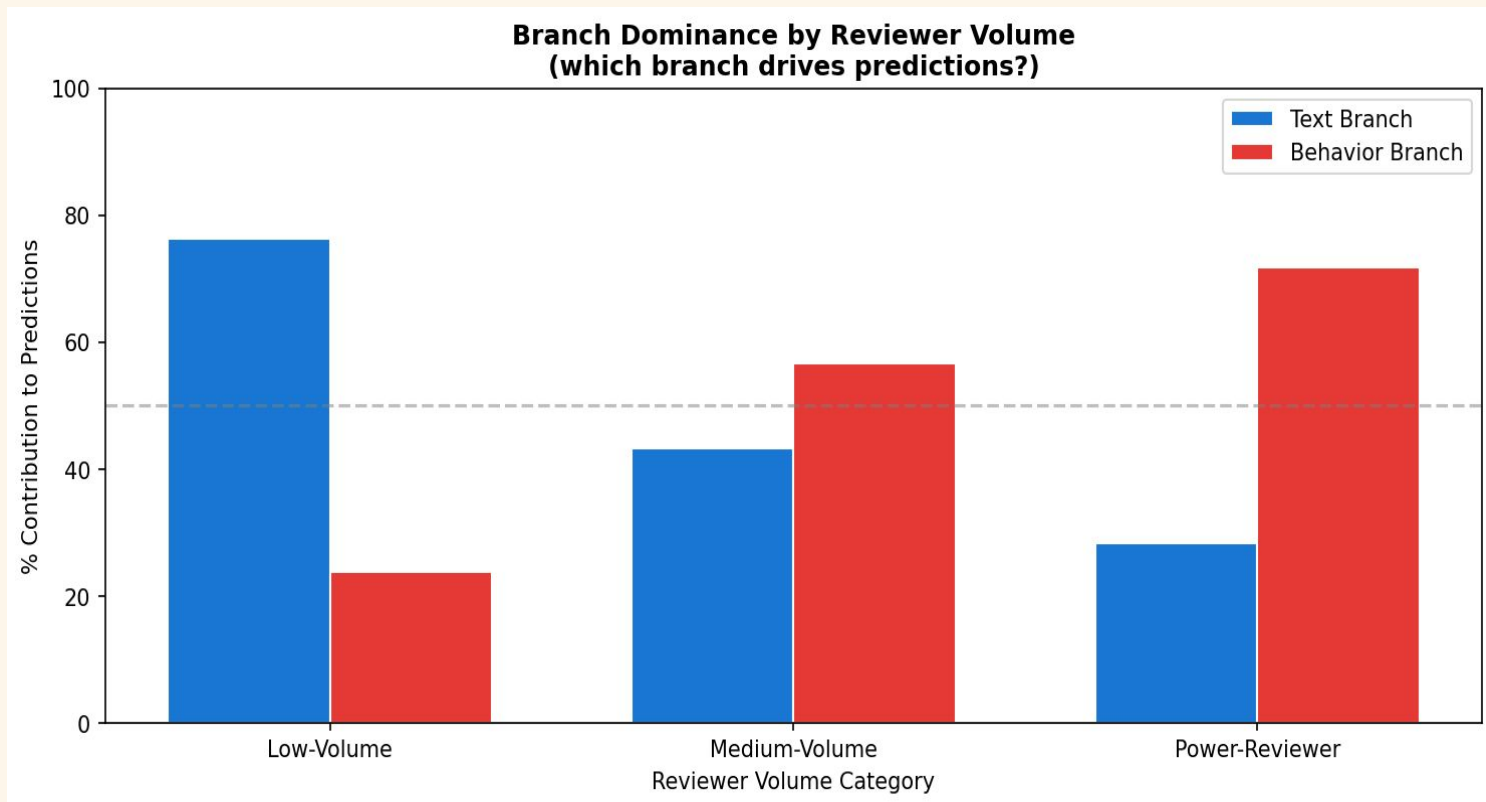


Figure 6. SHAP reviewer type analysis

Dynamic Attention Logic

New Users: The model prioritizes **Text (RoBERTa)** as behavior patterns are not yet established.

Power Users: Reliability shifts to **Behavior (Metadata)**, analyzing long-term patterns.

Result: Creates a robust system that is significantly harder for professional spammers to "game."

Limitations



Computational Cost

RoBERTa is a massive model. Running this in real-time for millions of reviews would require significant GPU power.



Class Imbalance

Only 14% of reviews are fake. Even with Focal Loss, the model still has to work much harder to find "needles in a haystack."



Temporal Shift

Spammers change tactics over time. A model trained on 2015 data might need re-training for 2024 slang or AI-generated spam.

Conclusion



Success

We successfully fused deep Transformer semantics with reviewer behavior.



Insight

Explainable AI (SHAP) is essential for building trust in automated moderation systems.



Impact

This architecture provides a robust, adaptable framework for protecting consumers from digital deception.

Future Work

- **Graph Integration:** Future versions could add a 3rd branch for **Social Graphs** (detecting rings of friends who all upvote each other).
- **Generative AI Detection:** Integrating specialized detectors for LLM-generated fake reviews (GPT-4 based spam).
- **Cross-Platform Learning:** Training on Yelp and then applying the knowledge to Amazon or TripAdvisor reviews.

Thank You 😊

Questions?